

Pakistan produces a great variety of delicious fruits. But these are grown in specific areas due to variation in climatic conditions; Balochistan and Khyber Pakhtunkhwa are the major fruit producing areas. These fruits include grapes, apple, pomegranates, plum, apricot, pears and cherry. In Sindh, few varieties of fruits like mango, dates, banana, watermelon and melon are produced. In the Punjab, mangoes, oranges, kino, musk melon, water melon and dates are grown. Dry fruits like almond, pistachio and walnut are grown in Balochistan and Khyber Pakhtunkhwa. Pakistan earns lot of foreign exchange by exporting fresh as well as dry fruits.

Agricultural Problems of Pakistan

Agriculture is the main occupation of the people of Pakistan. More than 70% of the people are involved in agriculture. In spite of the fact that a large number of cash crops and food crops are grown; the agricultural yield rate is very low. The reasons for this low production are as under:-

a. Low literacy rate

Literacy rate of the country is quite low. Majority of our farmers are not educated and do not have knowledge about the modern farming. They have little knowledge about the use of insecticides, selection of good quality seeds and proper use of fertilizer, with the result that their production per acre is far below the requirements of the country. They only believe in and use the traditional methods of cultivation which they learnt from their elders with the result that they get low yield from their land.

b. Increasing number of farmers

The number of people who depend on agriculture is increasing but the process of bringing more area under cultivation, is quite slow. This has resulted in decrease in per capita area under cultivation.

c. Non-mechanized cultivation

Our farmers are still using old wooden plough, animal dung fertilizers, uncertified local seeds and old methods of cultivation. This is the reason that per acre yield is not increasing in spite of the fact that our



farmers are hardworking and laborious. Mechanized cultivation is not followed. Tractor, tube well, natural fertilizer, certified quality seeds and disciplined sowing of seeds are the necessary elements of mechanized cultivation. Our farmer is hesitant to adopt mechanized farming either due to conservative ideas or due to lack of financial resources as also due to the small land holdings.

d. Limit of agricultural Land

Agricultural land of Pakistan is divided into two groups. The landlord group has vast areas of land but they do not cultivate themselves, hence vast areas of land remain uncultivated and barren. The second group has canal-irrigated areas of land which is not more than 12-15 acres per family/person. They cannot take on mechanized farming on small areas. Without mechanized farming production will be low. They divert their attention to other jobs. This results in low production.

e. Role of Agriculture Department

Role of Agriculture Department is not much effective because our farmers do not have faith in the advice of Agriculture Department. The farmers believe in their old methods and the experiences gained from their forefathers. On the other hand the staff of agriculture department is also lacking in performing an effective role. Absenteeism from duty, non-supply of advice at an appropriate time and laxity in educating the farmers are the reasons for the non-co-operation between the farmers and the department. As a result agricultural output suffers.

f. Soil erosion

Soil erosion is caused by rains and other natural agents, like wind storm, snowfall and earthquakes. They remove away fertile upper crust particles of earth. The result is low yield.



g. Water logging and salinity

Large canal areas in the province of Punjab and Sindh have been rendered useless due to waterlogging and salinity. The loss of fertile land causes serious setback to the fertility of the land under cultivation.

h. Fragmentation of landholdings

As a result of fragmentation, the fertile land is divided into small land holdings which restrict large scale production and better use of the land including mechanized farming and the use of modern equipment.

i. Insufficient means of transport

Villages are the main areas of agricultural production. But metalled roads are either not available or are in a bad shape, causing obstacle in quick transportation. A good quantity of production does not reach safely to the market. Farmers, therefore, contend themselves with small production of crops.

j. Difficulties in the marketing of crops

The farmers do not get full or proper price for their products due to different tactics and difficulties created by the middlemen or those who have influence over the marketing process. This makes the farmers disappointed who pay little attention to increase the production.

k. Unsuitable conditions

Due to the unsuitable living conditions in the countryside and non-availability of essential medical facilities, health of farmers and their families is affected. Their strength for more production is minimized. This also affects the production of crops.

Measures to solve Agriculture Problems

The government of Pakistan has taken a number of steps to solve the agriculture problems in the country. These steps are as under:



a. Educational facilities

Government has extended educational facilities to the farmers through agriculture department in the use of modern scientific methods of cultivation. An overall programme of adult education has been started in the rural areas. Radio and television broadcast and telecast programmes on agriculture are to educate the farmers about modern methods of cultivation. Pamphlets and leaflets have been published for creating awareness among the farmers about modern cultivation.

b. Provision of soft loans

The government has started schemes for soft-term loan facilities to the farmers for purchasing modern agriculture equipment, fertilizers, high yield variety seeds and insecticides. Loan for purchasing tractors and installation of tube wells are also provided. These loans are recovered in easy installments.

c. Means of Irrigation

Artificial means of irrigation are being made more effective for providing proper and timely water supply to the fields. Arrangements to eliminate waterlogging and salinity have been made. Such trees are being planted which have long roots and help in eliminating this disease. The canals and the water courses are being lined.

d. Introduction of Agriculture Vocations

In order to reduce pressure of population on the area under cultivation, a number of vocations related to agriculture are being promoted to associate people with these professions. These supplementary professions also provide an opportunity to the farmers to earn extra money through profitable use of their spare time.



e. Land Reforms

The government introduced land reforms in 1959, 1972 and 1977. Under these reforms the government fixed a ceiling on land-holdings of big landlords and distributed the surplus land among the landless farmers. The objectives of land reforms were to improve relations between the tenants and the land-owners, abolish the monopoly of big landlords and develop the system of agricultural productivity. But still there is a need to make more reforms so that uncultivated land is made productive."

f. Agriculture Institutions

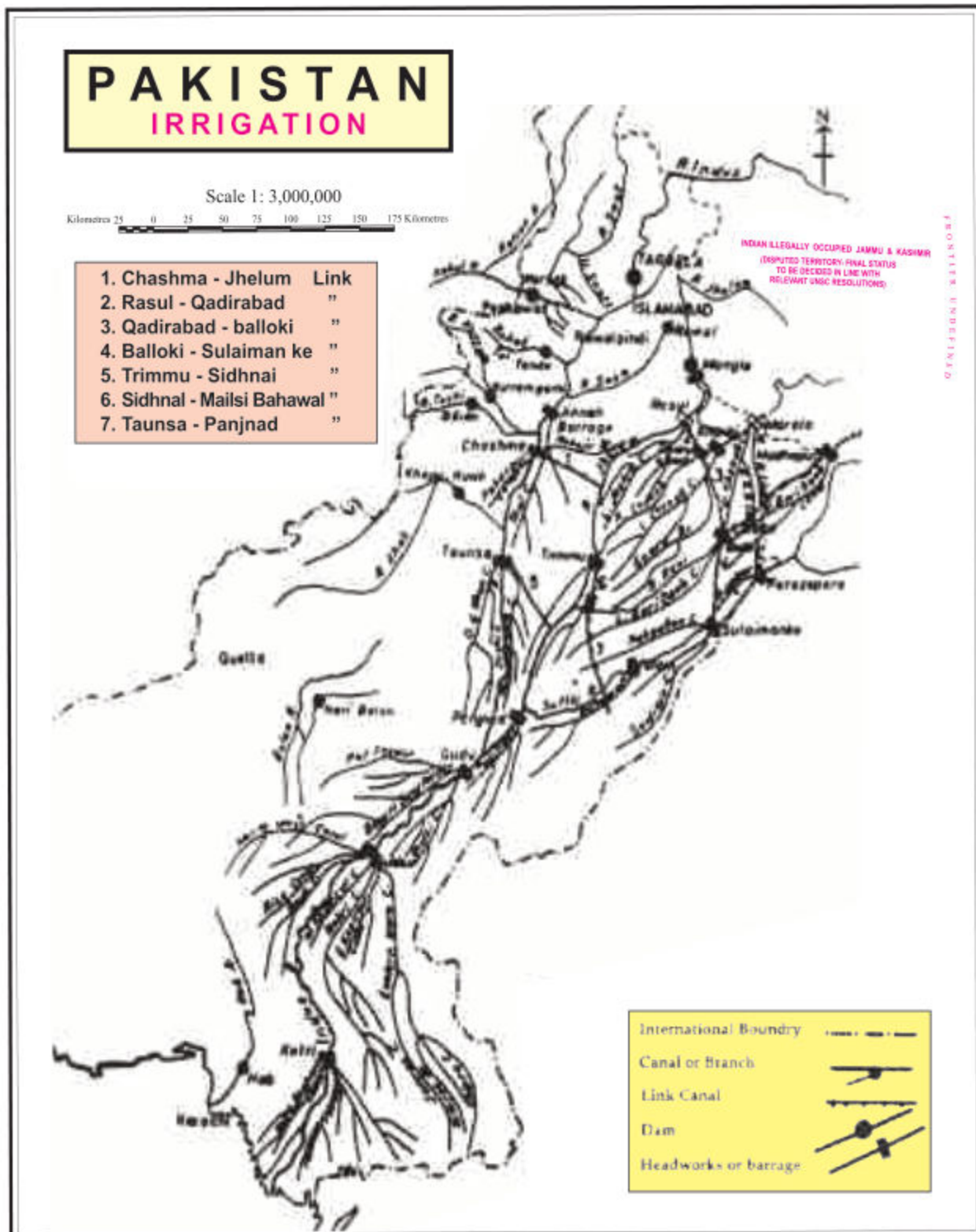
Government has set up agriculture institutions such as Agriculture Universities and the Colleges in the country. These institutions are providing education in different subjects of agriculture and are producing graduates in Agriculture Science. The government has established agricultural institutions like:-

- i. Agriculture University, Peshawar.
- ii. Arid Agriculture University, Rawalpindi.
- iii. Agriculture University, Faisalabad.
- iv. Agriculture University, Bahawalpur.
- v. Agriculture University, Tando Jam.
- vi. Agriculture College, Multan.
- vii. Agriculture Institute, Dokri, (Larkana).

The Agriculture Research Institutes in the country are engaged in finding the ways and means to increase agricultural production, develop high quality seeds and discover innovative methods to control plant diseases.

B) Irrigation

Pakistan is basically an agricultural country. Its 73% area is under cultivation which depends on irrigation through canals and other sources like tube wells, wells and Karez. In most parts of Pakistan annual rainfall is about 250 mm which is very low and of uncertain. Pakistan has the biggest canal system in the world. There is a wide spread network of canals in Punjab and Sindh through



head works, barrages and dams. There are few canals in Balochistan because there is no big river and the land surface is uneven. The Pat Feeder Canal takes its water from the Indus (in Sindh). The lands are irrigated by Karez also.

Canals are the important source of irrigation. River water is provided to distant areas through these canals to increase and sustain agriculture production. Pakistan has the largest system of canal-irrigation in the world. There are two types of canals. Perennial canals flow throughout the year and the non-perennial canals flow only during the rainy season or at the time of floods. The number of non-perennial canals is very small. There are 18 head works and barrages and 38 big canals in Pakistan. There are two large and a few small dams in Pakistan. The Mangla Dam on Jhelum River and Tarbela Dam on the Indus River are the two large-sized dams.

Source of Canal System of Pakistan

As a result of the partition of sub-continent, the province of Punjab was also divided into East and West Punjab. East Punjab went to India. The partition of Punjab led to the partition of the canal system of undivided Punjab. The head works of the Sutlej and the Ravi canals were in India. Some parts of Pakistan were irrigated by these canals. India stopped water of these canals. This created water dispute between India and Pakistan. In order to find out a permanent solution, World Bank in September 1960 engineered an agreement between the two countries. The agreement is known as 'Indus Basin Treaty'. The main features of this treaty are as under:-

- a. India got the rights of three Eastern rivers namely Ravi, Beas and Sutlej.
- b. Pakistan got exclusive rights of three Western rivers, namely, Indus, Jhelum and Chenab.
- c. Pakistan received water through the link canals into the canals coming out of the rivers, Sutlej and Ravi.
- d. Pakistan would construct two storage dams, five barrages and eight link canals to meet shortage of water due to loss of three eastern rivers which went to India.

Under this agreement, Mangla Dam and Tarbela Dam were constructed. These dams are the source of providing water to canals. The old canals were widened and barrages were expanded.

Now, there are 4 famous Dams in Pakistan namely Mangla Dam, Tarbela Dam, Warsak Dam and Ghazi Barotha Dam. The number of headworks is increased to 18 and that of big canals to 38.

C) Livestock

Livestock is an important section of agriculture in Pakistan, which accounts for 37.5% of the value of agriculture sector and contributes about 10% to the national output of Pakistan. It is a source of foreign exchange earning for Pakistan. Livestock includes cattle, buffaloes, sheep, goats, camels, horses, donkeys, mules and poultry. The livestock products are milk, beef, mutton, poultry meat, wool, hair, fats, blood, hides, skins, etc. Fish farms are also source of livestock. Some of the livestock like camels, horses, donkeys and mules are also used for transport, travel and for tilling the land.

In different parts of Pakistan, breeding of cattle is done on commercial basis. Dairy farms have been established in Punjab and Sindh. The Government has established cattle farms in Sindh at Malir; Mirpurkhas, Sakrand, Dadu and Tando Muhammad Khan. In Punjab, cattle farms have been established at Bahawalpur, Vehari, Khanewal, Dera Ghazi Khan and Sahiwal.

Fish farms are also gaining importance and a number of fish farms have been set up in Punjab, Sindh and Khyber Pakhtunkhwa. About 2 lac people are engaged in the fishing profession. Poultry is a big industry and has flourished throughout Pakistan. In Balochistan and Cholistan where rainfall is scanty rearing is common. Pakistan is among those countries where conditions of livestock preservation are not satisfactory. The main reasons are traditional methods of livestock rearing, lack of sufficient veterinary hospitals and shortage of trained staff in these hospitals. With proper planning, Pakistan can earn valuable foreign exchange from the export of livestock and poultry products.



4. Power Resources

Power resources include coal, mineral oil, natural gas, nuclear energy, wind power and solar energy. These resources are:

a. Hydro and Thermo Electricity (Hydel and Thermal Electricity)

Thermal and hydel (water) energy is commonly used in our country.

Electricity is generated through Thermal and hydro resources. Coal, oil and gas are used to produce thermal electricity. But we cannot depend upon these resources because in Pakistan coal, oil and gas deposits are limited. Thermal power station can be set up easily but its generation cost is very high. Therefore, it is an expensive venture. However, a thermal power station is being established near coal mines of Tharparkar. At present about 58% electricity is obtained through thermal resources. The thermal power houses have been set up at Faisalabad, Multan, Kot Addu, Rohri, Jamshoro, Hyderabad and Karachi.

The hydro-electricity is produced through water of the rivers. The hydro-electricity projects are located at Tarbela, Mangla, Warsak and Ghazi Brotha. About 42% of the electricity is produced through water resources. Although establishment of hydro-electric stations is an expensive business but its cost of production of electricity is quite less than the cost of thermal electricity. After the electricity is produced from the water of the rivers the water can again be used for irrigation. Hydro electricity generation does not create air pollution.

b. Atomic or Nuclear Energy

Another source of energy is atomic power or nuclear energy. Uranium, which is a heavy grey radioactive metallic element (symbol U), is used for the production of atomic energy. In Pakistan two atomic power houses are working. One is at Karachi, which started its work in 1971 and the second is at Chashma (Mianwali). The Chashma plant started producing electricity in 2002. Its production capacity is 300 MW. A third atomic power project is being constructed again at Chashma with the collaboration of China. Atomic power stations are preferred because they supply cheap electricity to consumers. Atomic energy is also used for peaceful purposes. For example, it is used for agricultural research. In this regard three centers at Faisalabad (Punjab), Peshawar Khyber Pakhtunkhwa and Tando jam (Sindh) have been set up for agriculture and food research besides the National Institute of Food and Agriculture (NIFA). Atomic energy is used for the treatment of cancer. Such institutes have been set up at Peshawar, Lahore, Karachi, Jamshoro, Larkana, Multan, Bahawalpur, Islamabad, Quetta, Faisalabad and Abbottabad.

c. Solar Energy

Coal, oil and gas are non-renewable deposits and are likely to be consumed relatively quickly due to their excessive use. But solar energy is such a treasure which cannot be consumed. It is almost free. About 200 million MW solar energy is received each day throughout the world from the sun. It is about

60, 000 times the total productive capacity of all the power houses in the world. Solar energy is obtained through a number of methods. It is stored in the solar cells and used in radio and small vehicles. By using big size mirrors in the solar boiler, rays of the sun are diverted towards the boiler; energy is produced and used in operating big machines. The solar panels seek heat from the sun. In Pakistan sun shines quite brightly for about 300 days in a year. The sun-light can be used for producing electricity in the rural areas for cooking food, running the tube wells and for obtaining the heat.

Presently, it is too expensive to set up a solar power house in Pakistan. Also, technical expertise is not available. However, a few small solar energy units are working at the following places:

- i. Khurkhira (Lasbela, Balochistan)
- ii. Malmari (Thatta, Sindh)
- iii. Dittal Khan Laghari (Tharparkar, Sindh)
- iv. Hoot (Multan)
- v. Nasirabad (Gilgit)

Presently, solar energy sector has not been given much attention because electricity needs are being met through other sources.



5. Human Resources

Pakistan ranks fifth among the most populous countries of the world. It is the second populous country in the Muslim World. When Pakistan came into being its population was about 33 million. Now, it has reached about 200 million. It has been estimated that less than 30% population of Pakistan is involved in active work. Out of this estimated figure 40% are associated with agriculture, 18% with industries and 40% are employed in other sectors. Out of this population, rural and urban ratio is 66.5% and 33.5% respectively. This working population is indeed the human resource of Pakistan. The economic, social and political advancement of the country depends upon the human resource.

Human resource is an important factor for development, but the more important is the physical health, mental health, education and skillfulness of this human resource which determines its utility and effectiveness. The effectiveness and productivity of human resource in Japan can be visualized that its human resource is 100 million but their total production is 5 times bigger than the total population of all the Muslim countries, which is 1000 million. In spite of the rich resources of Muslim countries their total national production is 1200 billion dollars

whereas Japan's production is 5500 billion dollars annually. Japan has no oil, gas or coal deposits. But, it has the high level of education and development in science and technology. The hard work of the people has made it a Prosperous country. On the contrary, Muslim countries pay little attention to the development of their human resource. In Pakistan, feudal rulers did not pay attention to raise the standard of human resource in the country by providing funds in education, science and technology sector. It is a recognized fact that development of human resource is the best investment Government of Pakistan has now started paying attention to science and technical education. Science and technical institutions are being set up. Their budget has been raised almost 5 times as compared to the past. It is around 5 billion rupees.

Interdependence of human and other resources

Human resource and other resources have their independent and separate value as well as utility. But these resources are interdependent. Human beings have a number of needs for living, which include necessities of life like food, clothing and shelter. Similarly, some comforts and luxuries are part of human needs but rank after the necessities of life. These needs can only be fulfilled with the help of other resources, which include agricultural and mineral resources. To discover and obtain benefit from these resources, human efforts play an important part. If human knowledge, expertise and labour are excluded, the fruits of other resources cannot be obtained. The natural resources of oil, gas, gold etc. have been converted into wealth and given value because man discovered them and used them for its benefits. On the other hand, human life without these resources would have been charmless and even survival of human beings would have been impossible. So human resources and other resources are interdependent.

6. Problems and prospects

The fast growing population of Pakistan is a major problem and a source of hardship for the nation. Our resources are limited and there is a mismatch between our population and the resources. An imbalance is being created between the population of rural and urban areas. Migration from rural to urban areas is our major problem. This migration is due to search for jobs and better amenities of life. But, unplanned activities, devoid of future requirements, have created manifold problems. On the one hand, villages are losing their natural charm and on the other hand, urban areas suffer from the problems like shortage of water, electricity, health, sanitation, housing etc., Pollution in the cities has

created diseases. Crime is increasing day by day. The life in rural areas as well as in urban areas is becoming difficult. Rural areas do not have the required facilities whereas in urban areas facilities are available but are not enough for the growing population.

The only remedy to the problem is that unplanned migration of population from rural areas to urban areas should be stopped by taking following actions:

- a. Education for All (EFA) programme started by the Government should be strengthened by opening more educational institutions to make the people literate.
- b. Facilities like health, electricity and water should be provided on urgent basis to the rural areas including provision for housing.
- c. Job opportunities should be created by opening institutions of different kinds in the rural areas.
- d. Law and order situation in the rural areas should be made better to protect people from anti-social elements.
- e. Different indigenous jobs should be encouraged through soft term loans.
- f. Those who seek government employment should initially be posted in the rural areas, at least for 3 to 5 years.

7. Moderation in life

Moderation in life means to live within the available means. There is a saying that excess of everything is bad. Moderation is a way of proper thinking, behaving and acting. A person can be called moderate who makes an account of himself or herself and then decides to follow a path for his future life. Those who do not moderate their lives are exposed to troubles and hardships. Moderation demands reasonable attitude towards actions in all matters, economic, social and political. Moderation brings peace and prosperity in the society. Over ambitiousness is the cause of all evil practices. By following a moderate line of action one would lead a comfortable life.

As a nation, Pakistanis are highly emotional. We either completely indulge or take no care at all, which results in total indifference towards our duties. This

has made the society backward. Our extreme passions and sentiments have labeled us as an emotional nation. The emotions, sometimes, may bring temporary success but in the long run prove counter productive.

It is well-known to all that living within one's resources is a sign of prosperity. Those who exercise self-control and restraint lead a happy life. Again an over indulgent nation is always in difficulties for their extreme attitudes and actions. Islam has, therefore, taught moderation in all walks of life and has always stressed on self-control.

EXERCISE

A. Answer the following questions:

1. Describe the importance of resources.
2. Name different kinds of resources.
3. What are the advantages of forests?
4. Name the mineral resources of Pakistan.
5. What are the agricultural problems of Pakistan?
6. What are the sources of agriculture in Pakistan?
7. Write a note on the Nuclear Energy in Pakistan.
8. How are human and other resources interdependent?
9. Write note on moderation in life.

B. Fill in the blanks with appropriate words:

- i. The people involved in a number of tasks are called _____.
- ii. The soil formed from the flooded layers of silt and loam is called _____.
- iii. Mountain forests are found in _____.
- iv. The natural gas was first found in Pakistan at a place called _____.
- v. Moderation in life means _____.

Chapter 06

INDUSTRIAL DEVELOPMENT IN PAKISTAN



1. Meaning of Industry

Industry is the work and process involved in the making of things in factories. It refers to manufacture of things through manual or mechanical methods. In broader terms, industry means preparation of commodities from raw materials which are of some utility to human beings. The industry has three fold advantages. They fulfill the necessities of human beings; add to their comforts through easy and quick preparation of things.

In the early period of civilization the industry was simple and of quite low level as compared to modern industry. Industry was started at a low level but gradually developed. Now industry is a big sector of production. Different industries have merged into single unit, reducing cost of production of commodities due to large scale output.

Meaning of National Development

National Development is a process of advancement in economic and social sectors. The resources have been discovered and exploited to the maximum utility of the masses. The people have a standard of living, which guarantees economic and social protection. In other words, it can be said that national development reflects strength of human and natural resources, which make the life rich and easy. Industry is one of the factors necessary for national development.

2. Industry and its importance for development

Industry is an-essential component for the development of a country. With the discovery of agricultural and natural resources it had become necessary to invent machines to get optimum benefit of the resources. Raw materials are converted into durable goods and help in production of food for human beings. Industry is important due to the following reasons:

- I. It is the source of economic development of a country. Those countries are considered developed which are developed in industry.

- ii. Commodities are manufactured on large scale to fulfill local and national needs; and are also exported to other countries to meet their needs. Valuable foreign exchange is also earned.
- iii. Value of raw materials is raised by manufacturing them into finished goods. Cotton as a raw material has less value as compared to cotton yarn or cotton cloth.
- iv. People get employment according to their skill or expertise and earn livelihood for themselves.
- v. Industry has added to the comfort of human beings. Radio, TV, Air conditioner, refrigerator and many other goods have added to the comfort of the people.
- vi. Industry makes the country economically strong. Strong economy helps in political and strategic stability.
- vii. Country becomes self-reliant and prosperous. Standard of living is improved.

Pakistan inherited such territories, which were industrially backward. Non Muslims were big capitalists and did not establish industry in the Muslim majority area in spite of the fact that in those areas raw material and cheap labour was available. For example, East Bengal produced 50% of the Jute in the world but no jute industry was set up, whereas in Hindu majority area of West Bengal about 100 jute mills were set up on the bank of river Hugly (or Hooghly). Cotton was grown in West Punjab (Pakistan) but big factories were set up in Bombay and Ahmedabad. At the time of independence seven factories of cotton, sugar and cement existed in Pakistan. So in reality, Pakistan started its journey with a weak industrial base.

Keeping in view the backwardness in the industrial field, the government of Pakistan started its efforts to provide an industrial base to the country. In 1948, an Industrial Policy was announced which encouraged private investment in the industrial field. In 1962 Pakistan Industrial Development Corporation (P.I.D.C) was set up by the government and large industries were set up with state investment. But in 1972, ten categories of factories were nationalized by the government. This action of the then government created uncertainty in the private investors and they hesitated to invest in the industrial sector. Presently the government is trying to privatise the industries, besides allowing the private sector to set up new industrial units. Foreign investors have also been invited and encouraged to invest in Pakistan.



3. Industries

In Pakistan industry can be categorised as under:-

- i. Cottage and small industry
- ii. Heavy industry
- iii. Defence industry

i. Cottage and Small Industry

Cottage and small industries are very important as they provide large employment at local level. They can be established with small investment and simple administrative set up. If properly organized they can be a good source of reducing migration of rural population to urban areas. Different varieties of consumer goods are prepared by these industries in different parts of the country. In Pakistan, role of cottage and small industries is very significant. Thousands of people are engaged in different trades and are contributing in the local as well as national economy. The cottage and small industries of Pakistan are as under:-

a. Carpet Industry

Raw material for carpet making is available in Pakistan. Carpets are prepared in different parts of the country. The important centers of carpet making are Lahore, Sheikhupura, Faisalabad, Multan and Jhang in the province of Punjab. In Sindh carpet centers are located in Jacobabad, Sukkur, Khairpur, Mirpurkhas, Tharparkar, Umerkot, Hyderabad and Karachi. In Khyber Pakhtunkhwa, carpet making is done in Peshawar on a large scale and it is a big profession at local level similarly, in Quetta (Balochistan) a large number of people make carpets. Beautiful carpets are prepared in Pakistan. These are popular in the foreign countries. This small industry is also earning valuable foreign exchange. Carpets are prepared by using wool and synthetic materials.

b. Cotton Handlooms

This is an important industry among the cottage and small industries. This industry includes handlooms whose network is



spread in Punjab and Sindh. The number of these handlooms is in thousands. These provide a big source of local employment. These cotton handlooms produce a variety of beautiful blankets (khes or cotton shawls), bed sheets, cotton rugs, etc. In Punjab, center of cotton handlooms are at Faisalabad, Multan, Lahore, Gujranwala, Sargodha and Sialkot. Hyderabad and Sukkur are the centers of cotton handlooms in Sindh.

c. Leather Tanning

Leather manufacturing is an important industry of Pakistan. Varieties of shoes suitcases, vases, handbags and other leather items are prepared in different parts of the country. Leather items are manufactured at Karachi, Hyderabad (Sindh), Lahore, Kasur, Sialkot, Gujranwala, Sheikhupura and Multan (Punjab), Peshawar and Swat Khyber Pakhtunkhwa. Leather is tanned in Balochistan. A large number of people are engaged in this industry.

d. Sports Goods

The raw material required for the production of sports goods is available in abundance in certain parts of Pakistan. Soft timber and leather is required to make sports goods. Sports goods are prepared at Sialkot, and Lahore. These goods are also a source of foreign exchange earning Hockey, cricket, bat and ball, football and rackets made in Pakistan are very popular in foreign countries.

e. Cutlery

Different kinds of cutlery are produced at Wazirabad, Sialkot, Gujranwala, Gujrat and Lahore in Punjab. These are a source of earning big foreign exchange for the country.

f. Embroidery and needle work

Embroidery and needle work is the pride of Pakistan. It is a popular skill in Pakistan. Flower embroidery through silk thread is very famous and is an attraction to the people. This is done in different cities of Punjab, Sindh and Balochistan. The work of mirror ornamentation on cloth with silk thread, known as Balochi art is done on large scale in Sindh and Balochistan. The important centers of embroidery are at Larkana, Dadu, Shikarpur,

Nawabshah and Khuzdar. It is a popular profession of a number of families in Sindh and Balochistan. The work of "Salma Sitara" is done in big cities of Punjab and Sindh. Flower embroidery, leather embroidery and Zari work (golden embroidery) is done on cloth in Peshawar, D.I.Khan, Kohat and Nowshera in Khyber Pakhtunkhwa.

ii. Heavy Industry

a. Textile Industry

This industry is the backbone of Pakistan's economy. There are large numbers of big and small textile mills in Pakistan. Fine varieties of cloth are manufactured in these mills. Pakistan is self-sufficient in cotton cloth. Each year millions of foreign exchange is earned through the export of cotton cloth and yarn. Main centres of textile industry are Faisalabad, Lahore and Multan in Punjab, Karachi and Hyderabad in Sindh. In Khyber Pakhtunkhwa these centres are at Peshawar, D.I. Khan, Nowshera, Bannu, Haripur and Swat. Uthal and Quetta are the two textile centres in Balochistan.

About 50% of industrial labour is employed in the textile industry. The present number of textile mills in the country is 500 as compared to just 3 at the time of independence in 1947.

Pakistan has also a woollen textile industry but it is not as rich as cotton textile. The reason is that wool in Pakistan is not of high quality. Most of our wool is used for carpet making. However, major woollen textile centres are at Karachi (Sindh), Lawrencepur, near Rawalpindi, Lahore, Quaidabad (Punjab), Harnai, Mastung (Balochistan), and Bannu and Nowshera Khyber Pakhtunkhwa, where woollen yarn blankets and woollen cloth is manufactured. At present there are 70 woollen mills in the country. The Lawrencepur and Karachi Woollen mills produce high quality woollen cloth.

Pakistan has also a silk textile industry. Two types of silk are used for manufacturing silk cloth. One type is of natural silk obtained from silk worms and the other is the artificial silk called synthetic silk. Since natural silk has become rare and most expensive, synthetic silk has gained popularity. There is one Silk factory at Kala Shah Kaku near Lahore, preparing synthetic silk. This is known as rayon, silk yarn. Raw silk is imported from other





countries. Karachi is the biggest center of silken textile industry. Silk textile is also manufactured at Faisalabad, Lahore, Multan, Gujranwala, Peshawar, Swat, Sukkur and Hyderabad.

b. Sugar Industry

It is one of the biggest industries in the country. Pakistan started with two sugar factories at Rahwali near Gujranwala (Punjab) and Takht Bhai Khyber Pakhtunkhwa in 1947. Sugar is made out of sugarcane, which is cultivated in large quantity in the three provinces, namely, Punjab, Sindh and Khyber Pakhtunkhwa. Therefore, government decided to set up sugar factories in the areas Sugarcane is cultivated. There are 78 Sugar mills in country (40 in Punjab, 32 in Sindh, 6 in Khyber Pakhtunkhwa) with capacity to produce 5 million tonnes of refined sugar. Pakistan is not only self-sufficient in the production of sugar but also earns valuable foreign exchange through export of sugar. Pakistan's sugar is of best of quality.



c. Cement Industry

Limestone and gypsum is used in the manufacturing of cement. Fortunately Pakistan has large deposits of both limestone and gypsum. Therefore, a number of big cement factories have been established in public and private sector. The public sector factories are under the administrative control of Pakistan State Cement Corporation (PSCC). Cement factories have been established in the following cities of Pakistan:

Federal Area	:	Islamabad
Punjab	:	Dandot, Wah, Daud Khel, Rawalpindi and Dera Ghazi Khan
Sindh	:	Karachi, Hyderabad, Thatta, Nooriabad and Rohri
Khyber Pakhtunkhwa	:	Kohat, Haripur and Nowshera.
Balochistan	:	Darwaza and Gadani

Pakistan is almost self-sufficient in the production of cement. At present, there are 25 cement factories in Pakistan as compared to only one at the time of independence in 1947. The installed capacity is 17.7 million tonnes. There are 21 cement factories in the Private sector and 4 in Public Sector.



d. Vegetable Ghee and Cooking Oil Industry

This industry, in the beginning was established in the private sector but it was nationalized in 1973. Out of 26 factories, 23 factories were nationalized and put under the control of Ghee Corporation of Pakistan (GCP). Raw material for vegetable ghee industry's imported because local raw material is not sufficient. These are located at Karachi, Hyderabad, Sukkur and Shaheed Benazir Abad in Sindh, Rahim Yar Khan, Bahawalpur, Lahore, Multan, Faisalabad and Rawalpindi in Punjab. Nowshera, Haripur and Dargai in Khyber Pakhtunkhwa and in Dera Murad Jamali and Quetta in Balochistan. There are a few factories in the capital of Pakistan, Islamabad. The industry is now entirely in the private sector after privatisation. It comprises 160 units in organized and unorganized sectors. The installed capacity is 2.7 Million tonnes.



e. Chemical Fertilizer Industry

This industry is meeting the needs of chemical fertilizer necessary for agricultural production. Different kinds of fertilizers are produced by the fertilizer Industries established at different places. Pakistan is producing surplus chemical fertilizer which is exported to other countries. There are 10 fertilizer units in the country (five in Punjab, three in Sindh and two in Khyber Pakhtunkhwa) the total installed capacity is 5.6 million tonnes.

iii. Defence Industry

Defence industry is also under the category of heavy industry. This industry includes iron and steel mills, heavy mechanical complex and ship-building.



a. Iron and Steel Industry

Iron and steel industry depends on iron ore. The iron ore deposits are at Kala Bagh, Makarwal in Punjab, Langrial in Khyber Pakhtunkhwa, and Khuzdar, Ziarat, Chil Ghazi and Nokkundi in Balochistan. But these reserves are not enough for the country and hence we have to depend on imported iron ore and steel. Nevertheless, two large industrial units have been established to produce iron and steel and its products. These industries are:



i. Pakistan Steel Mill, Karachi

This mill is located about 40 kilometers away from Karachi near Port Qasim. It was built with the co-operation of Russia (former U.S.S.R) in 1973. It manufactures pig iron, iron sheet, coal tar etc. Thousands of people have been employed in this mill and lot of foreign exchange has also been saved.

ii. Heavy Mechanical Complex, Taxila

Heavy Mechanical Complex, Taxila was established with the help of China in 1968. It meets the requirements of railway automobiles, sugar mills, cement factories, textile machines, fertilizer factories, trucks and spare parts. With the establishment of Taxila complex our imports for steel and machines have been reduced. A lot of foreign exchange is saved, that would have been spent on imports of different kinds of machines and spare-parts.

b. Shipbuilding Industry

In 1956 Karachi Shipyard and Engineering Works was established at Karachi. Various types of ships are built there. This shipyard is also used for repairing of ships. Apart from manufacturing local ships, repairing of fishing boats, tugs and ships of other countries is also done in this shipyard. Shipbuilding industry has developed over the years and it is now in a position to cater to the needs of our country.

c. Arms and Ammunition Industry

The first arms and ammunition factory was built at Wah near Rawalpindi. The air defence material is manufactured at Havelian in Khyber Pakhtunkhwa and in Kamrah (Punjab). Some defence machines are manufactured in Machine Tool Factory at Landhi, Karachi. Similarly, missiles are also prepared in Pakistan. Pakistan has become almost self-sufficient in the production of traditional as well as sophisticated arms and ammunition.

4.

Means of Transportation and Communication

All sources, which help people, travel from one place to another and help in transportation of goods are called means of communication. These means are

old as well as modern. The old means of communication are the roads and the sea journey. For road travel simple carts were used, driven by the animals like horse's, donkeys and oxen. People also used to travel on foot. The roads were not much comfortable. It took lot of time to reach from one place to another, sea boats and small ships traveling with the direction of winds were used which were not much safer.

With the development of science and technology, means of communications have been improved. Roads have been constructed, metalled with concrete and bitumen (coaltar). Motor vehicles have taken the place of carts and animals. Modern ships have taken the place of boats. Travel is done by air also. All these means of communication have their own importance.

Means of communication are very important for the development of the country. They help in economic development. The material is carried to factories and manufactured goods are taken to the market. They reduce unemployment as people can seek employment at distant places and also travel frequently. In brief means of communication help to:-

- i. Develop both agriculture and industry of the country.
- ii. Promote national and international trade.
- iii. Promote unity and national solidarity as people travel to different parts of the country.
- iv. Strengthen defence of the country through quick mobilization of armed forces.
- v. Promote knowledge and art and make their benefits available to other parts of the country.
- vi. Help in maintaining law and order in the country and also provide quick relief in case of catastrophes like floods, earthquakes, fire, etc.

Pakistan has all the three means of communication, namely land (roads), sea and air. Let us take an account of our means of communication.

Roads:

Roads are an important source of communication and transport in Pakistan. Roads link different cities with each other. Roads are used for travel

and for transportation of goods. There are metalled roads as well as kutcha (non-metalled) roads in Pakistan. Kutcha roads are not easy to travel during bad weather.

Pakistan has a network of roads covering 259758 kilometers, including 162879 high types and 96849 low types of roads. The important roads of Pakistan are as under:

a. National Highway

This is the oldest as well as most important highway of Pakistan. Its length is 1735 kilometers. It starts from Karachi and goes to Peshawar and Torkham. It connects different cities situated on this road like Hyderabad, Nawabshah, Sukkur, Bahawalpur, Multan, Lahore, Rawalpindi and Peshawar. Each of these cities are further linked with minor cities through small roads. A part of this road between Lahore and Peshawar is called Grand Trunk Road (G.T.Road).

b. Karachi - Quetta Highway Via Khuzdar

It joins Karachi with far-flung areas of Balochistan. It is a good means of transport. The main cities on this road are Lasbela, Wadh, Khuzdar, Kalat and Quetta. It is a metalled road. Its length is about 816 kilometers.

c. Karachi Quetta Highway via Jacobabad

This is metalled road, which connects Karachi with Quetta goes through Karachi, Kotri, Dadu, Larkana, Jacobabad, Sibi and Quetta. Its length is 762 kilometers.

d. Quetta Peshawar Highway

This highway connects Peshawar Khyber Pakhtunkhwa with Quetta (Balochistan). It passes through Quetta, Muslim bagh, Qila Saifullah, Zhob, Bannu, Kohat and Peshawar. Its length is 535 kilometers.

e. Quetta Multan Highway via Loralai

This metalled road connects Multan with Quetta. This road passes through Quetta, Muslambagh, Qila Saifullah, Loralai, Fort Munro, Dera

Ghazi Khan, Muzaffargarh and Multan. At Multan it joins the National Highway, which goes from Karachi to Lahore.

f. Attock Multan Highway

It connects Attock with Multan after passing through Multan, Bhakkar, Mianwali and Attock.

g. Regional Co-operation for Development (RCD) Highway

It connects Pakistan with Iran and Turkey. It starts from Karachi and passes through Lasbela, Wadh, Khuzdar, Kalat, Noshki, Nokkundi, Taftan to Iran and Turkey.

h. Peshawar Karachi Highway (Indus Highway)

Along the right bank of river Indus this road connects Peshawar with Karachi. It is the second largest highway of Pakistan. This road goes from Peshawar to Karachi via Kohat, Bannu, Dera Ismail Khan, Dera Ghazi Khan, Kashmore, Shikarpur, Larkana, Dadu and Sehwan.

i. Karachi Hyderabad Super Highway

It is wide and double highway. Its length is about 170 kilometers. It has reduced the distance between Hyderabad and Karachi because distance between Hyderabad and Karachi via National Highway is much longer. Therefore, people traveling in between Karachi and Hyderabad prefer to travel through Super Highway. Now this Highway have been Converted in to Motorway.

j. Lahore Islamabad Motorway

A road from Lahore to Islamabad, generally called Motorway, has been constructed with billions of rupees. Its length is 367 kilometers. It is of International standard. It has 3 lanes on either side. The important towns on the road starting from Lahore are Pindi Bhatian, Salim, Kot Momin, Bhera, Kallar Kahar, Balkasar, Chakri and Islamabad. At Pindi Bhatian it connects Faisalabad through another road. The motorway has been extended to Peshawar having its length 155 kilometers.

The government of Pakistan has set up a National Highway Authority (NHA) with the important task of improving the transportation

and communication network. It has started some new projects for the construction of good roads in different parts of the country to link them by reducing the distances.

Railways

The other means of communication in Pakistan is the Railways. After the establishment of Pakistan, Railways played a significant role in providing travel and transportation facilities. It is comparatively safer to travel by railway than by road and quicker as well. The network of Pakistan Railways comprises 7791 route kilometers, 815 stations and 46 train halts. Its major assets include 580 diesel and electric locomotives, 2275 passenger coaches and 21732 freight wagons. Since these assets have become quite old, government is paying attention to their rehabilitation and replacement.

A number of trains run on these routes. These are Tazgam, Tezro, Awami Express, Super Express, Shalimar Express, Chenab Express and Khyber Mail, etc. Some new trains have been started namely, Karakoram, Night Coach, Jaffar-Jamali Express (Rawalpindi-Quetta Express). The important railway routes are:

i. Peshawar Karachi via Rawalpindi, Lahore and Rohri

It is one of the longest railway tracks in the country. It starts from Peshawar and ends at Karachi. Its total length is 1672 kilometers. This railway track passes through a number of big cities, which are Peshawar, Nowshera, Rawalpindi, Jhelum, Gujrat, Gujranwala, Lahore, Sahiwal, Khanewal, Multan, Bahawalpur, Rahim Yar Khan, Nawabshah, Shahdadpur, Hyderabad, Kotri, Jungshahi and Karachi.

ii. Quetta Zahidan

A railway track from Quetta runs through Spezand, Noshki and Dalbandin and finally reaches Zahidan which is a border town of Iran.

iii. Rohri Quetta

This railway track coming from Peshawar to Karachi is diverted at Rohri railway Junction. From Rohri it goes to Quetta via Sukkur, Habibkot, Shikarpur, Jacobabad, Sibi and Quetta.

iv. Multan to Jacobabad via Dera Ghazi Khan

The railway which goes to Karachi from Peshawar is diverted from Multan and goes to Jacobabad via Muzaffargarh, Kot Addu, Dera Ghazi Khan and Kashmore. At Jacobabad it joins the Karachi Quetta railway line.

v. Quetta Zhob

A railway line travels from Quetta to Zhob through Bostan, Muslim Bagh and Qila Saifullah.

vi. Karachi Faisalabad

The railway track which starts from Karachi to Lahore and Peshawar is diverted at Khanewal and goes to Faisalabad.

vii. Rawalpindi Faisalabad via Wazirabad

This railway line goes from Rawalpindi to Faisalabad via Jhelum, Gujrat and Wazirabad. At Wazirabad through another track it goes to Faisalabad passing through Mandi Bahauddin and Sargodha.

viii. Peshawar Karachi via Rawalpindi, Faisalabad

The railway line starting from Rawalpindi is diverted at Wazirabad and goes further to Faisalabad and Khanewal / Lodhran. From here it joins the main track between Karachi Peshawar.

There are other branch railway lines between Multan and Rawalpindi and from Rawalpindi to Kohat. Some railway lines have been closed because these were a burden on the resources of railway.

Airways

Access to some parts of Pakistan is possible by air travel. For example, travel to Northern areas is easier by air as compared to roads. Similarly, some parts of Balochistan like Pasni, Gwadar and Turbat are difficult to travel by road. Air travel is a quick means of travel. Distances are covered in hours. Airways in

Pakistan were established in 1955, known as Pakistan International Airline (PIA). It has progressed rapidly and has established a network of air travel inside and outside the country. The main cities which are connected by air, inside the country are Karachi, Sukkur, Nawabshah, Moen-jo-Daro, Lahore, Islamabad, Peshawar, Quetta, Multan, Faisalabad, Sargodha, Mianwali, Dera Ismail Khan, Gilgit, Skardu, Chitral, Pasni, Gwadar and Turbat. New airports are being constructed and old airport are also being improved and expanded. There are 44 airports in Pakistan out of which 37 are operating with a fleet of 40 aeroplanes. At present four airlines in Pakistan namely, PIA, Aero Asia, Blue Airline and Shaheen Airline are extending full services to passengers.

Pakistan International flights network is linked with all important countries of the world. These countries are USA, UK, France, Canada, Germany, Japan, Emirates, China, India, Malaysia, Singapore and a number of other countries.

Waterways or Sea Routes

In Pakistan the use of waterways for transportation is very rare. In our rivers, the flow of water varies with the season. This is the reason that they cannot be used as permanent means of transport. Nowadays people don't prefer to travel by sea. The sea route is more popular in trade. Pakistan developed two seaports at Karachi and Bin Qasim. A third seaport is under construction at Gwadar. Pakistan's exports and imports are through sea routes. Pakistan has developed its own merchant fleet in order to reduce dependence on foreign companies. A Pakistan National Shipping Corporation (PNSC) was established in 1963. This Corporation acquired many new ships and repaired and reconditioned the old ones. Trade, economic and commercial links have been established with USA, UK, China, Japan, Australia, Hong Kong, Singapore, the Gulf countries as well as other countries.

5. Trade and Commerce

Trade and Commerce are important for the economic development of a country. Agricultural and Industrial development flourished with the help of trade and commerce. There are two types of trade one is internal or domestic and the other is external or International trade.

Pakistan's Internal Trade

Trade is an important occupation of the people of Pakistan. Inland trade is the source of distribution of goods, their sale and purchase within the country. Money remains in circulation with the help of internal trade. In Pakistan, huge trade activities run throughout the year and in all seasons. Punjab supplies wheat, rice, cotton, cloth, sports goods, stationery, machinery, cement and other products to other provinces. From Sindh cotton cloth, silk cloth, rubber goods are sent to Balochistan and Khyber Pakhtunkhwa, and the Punjab. Blochistan supplies dry and fresh fruits like plum, apricot, grapes, Pomegranates and apples to other provinces. Inter provincial trade has flourished at a large scale and articles like tobacco, cigarettes, timber and items made from sugarcane are supplied to the areas where these are required. Internal trade has become source of employment, welfare and prosperity of the people of Pakistan. Internal trade covers retail as well as wholesale business. Pakistan's big trade centers are Karachi, Hyderabad, Quetta, Multan, Lahore, Faisalabad, Gujranwala, Sialkot, Rawalpindi and Peshawar.

External Trade

No country of the world is self-sufficient in all the necessities of life. The deficient needs are met through their purchase from other countries and surplus commodities are sold to other countries. This trade is called external trade. Different countries are famous for different products. For example Japan's famous for electronic goods and motor vehicles. Pakistan is famous for carpet making, cotton cloth, leather goods etc. USA is known for heavy Industry and arms and ammunition. So, surplus goods are exported on demand and in return deficit goods are imported.

Pakistan's exports are cotton, cotton cloth, rice, sugar, carpets, fish, surgical instruments, fruits and vegetables. Pakistan is also exporting its defence products to some countries. It imports aircrafts, heavy machines of different kinds. Chemicals, medicines, iron ore, edible oil, tea, petroleum, electronics and scientific equipment's. Pakistan's main trade partners are USA, UK, European Union and Gulf countries, Saudi Arabia, Japan, China, Sri Lanka and Bangladesh.

Pakistan's imports exceed its exports. Pakistan runs a deficit in trade with the result that it is under huge annual debit of 3 billion dollars. Pakistan has to balance its international trade with more exports. This will be possible by improving standard of its goods and bringing the prices of its goods at the competitive level with other countries of the world.

E - Commerce

It is the abbreviation of Electronic Commerce. It means trade with the help of computers and internet. Through electronic business, deals are finalized accurately within little time. Electronic commerce is a branch of Information Technology. It helps in maintaining an updated correct record of trade and business. It also facilitates our imports and exports. In Pakistan the use of E-commerce is gaining ground. Through E-commerce contacts can be established with leading business organizations in the world through their websites and find out the details concerning their commodities, their prices quantity and time limit for supply of their goods. Payments can be made through internet. The shares of stock exchange of New York, Singapore, Hong Kong, London, etc. can be purchased or sold while sitting at home in Pakistan, provided an account has been maintained with them. E-Commerce has made the trade and business quicker, easier and better.



How Industrial Development can be increased?

Our industries are suffering from a number of problems. Knowing these problems will help us in finding their solution and increasing the pace of industrial development. These problems are:-

- i. There is shortage of skilled workers for the industry.
- ii. Absence of the institutions which can help in maintaining standard of goods with quality control.
- iii. Inconsistent policies on industrial development like nationalization and violation of rules in the name of industrial development.

- iv. Deteriorating law and order situation in the country has discouraged investment in the industrial sector.
- v. Energy crisis like uncertain prices of electricity and fuel.
- vi. Lack of proper marketing.
- vii. Lack of efforts for competitive prices of commodities in the international market.
- viii. Negative attitude of trade unions resulting in low production.
- ix. Lust for undue profit by the industrialists, businessmen and importers.

Remedial Measures

Following remedial steps will, help in increasing the industrial development:

- i. Law and order situation in the country should be improved enabling the investors to invest money without the fear of loss of life or assets.
- ii. Rules and regulations for trade should be made easier and simplified. Red Tapism should be eliminated.
- iii. A sense of dignity of labour should be created while training the staff in order to create sense of devotion to work and doing hard work for more production.
- iv. A strict system of quality control should be set up. There should not be compromise on the quality and standard of the finished products.
- v. Quality of education in marketing and business administration should be made better.
- vi. Industrial policies should be well-defined and stable.
- vii. Smuggling of foreign goods should be checked strictly.
- viii. Government should provide incentives to the industrialists in the form of relaxation in taxes, subsidies on quality production and training of the workers.
- ix. The working conditions of the workers are improved and their wages be increased.